

## UEI805: ENVIRONMENTAL INSTRUMENTATION

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| <b>L</b> | <b>T</b> | <b>P</b> | <b>Cr</b>  |
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**Course Objectives:** To understand the concepts of pollution monitoring, to enable select, design and configure pollution monitoring instruments

**Air Pollution:** Impact of man on the environment: An overview of air pollution sources and effects, Metrological aspect of air pollutant dispersion, Air pollution sampling and measurement, Air pollution control methods and equipment, Air sampling techniques, soil pollution and its effects, Gas analyzer, Gas chromatography, Control of specific gaseous pollutants, Measurement of automobile pollution, Smoke level meter, CO/HC analyzer.

**Water pollution:** Sources And classification of water pollution, Waste water sampling and analysis, Waste water sampling techniques and analyzers: Gravimetric, Volumetric, Calorimetric, Potentiometric, Flame photometry, Atomic absorption spectroscopy, Ion chromatography, Instruments used in waste water treatment and control, Latest methods of waste water treatment plants.

**Pollution Management:** Management of radioactive pollutants, Noise level measurement techniques, Noise pollution and its effects. Instrumentation techniques in solid waste management, social and political involvement in the pollution management system.

**Course Learning Outcomes (CLO):** After the completion of the course student will be able to:

1. Identify and analyze the sources and effects of air and water pollutants.
2. Describe air pollution sampling and measurement techniques.
3. Explain water sampling and analysis techniques.
4. Analyze solid waste management practices and noise level measurement techniques.

### **Reference Books:**

1. Bhatia, H.S., *A Text Book in Environmental Pollution and control*, Galgotia Publication (1998).
2. Dhameja, S.K., *Environmental Engineering and Management*, S.K Kataria (2000).
3. Rao, M.N. and Rao, H.V., *Air Pollution*, Tata McGraw Hill (2004).
4. Rao. C.S., *Environmental Pollution Control*, New Age International (P) Limited, Publishers (2006) 2<sup>nd</sup> ed.

### **Evaluation Scheme:**

| S.No. | Evaluation Elements                         | Weightage(%) |
|-------|---------------------------------------------|--------------|
| 1.    | MST+EST                                     | 70           |
| 2.    | Sessionals(may include assignments/quizzes) | 30           |